Thermodynamic description of the Al–Si system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Al–Si binary system as calculated with pycalphad from the SGTE dataset AlSi-12Tan.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Al) + (Si), lies at 850 K.

1. Introduction

The Al–Si system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as simple eutectic. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of Y. Tang, Y. Du, L. Zhang, X. Yuan, G. Kaptay, Thermochim. Acta, 527, 131-42(2012). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

The melting point of Al reproduced by the dataset is 935 K. Pure Si melts at 1687 K.

2. Phase diagram and invariant reactions

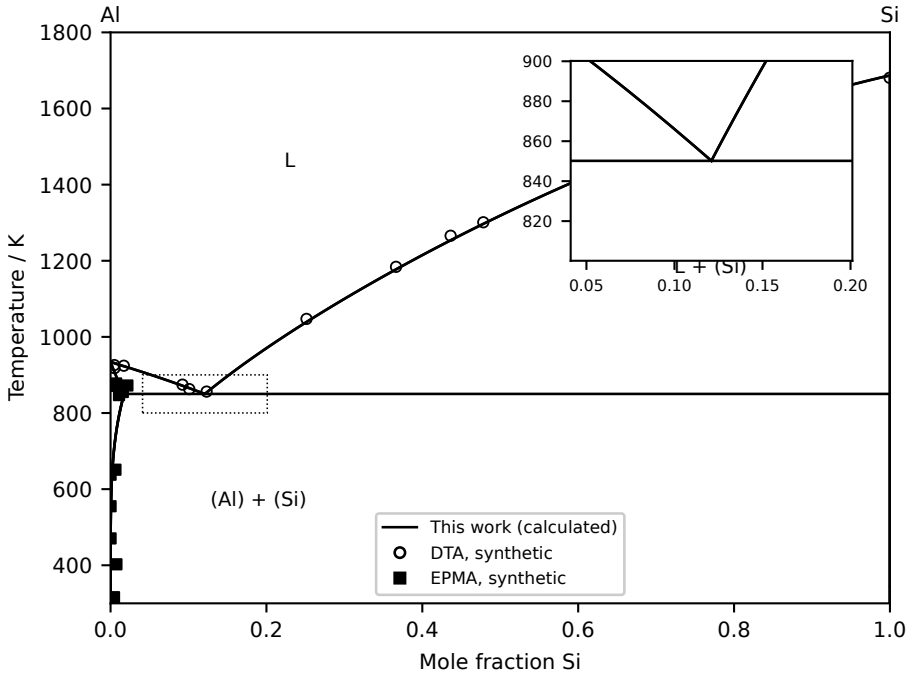


Fig. 1. Calculated Al–Si phase diagram with synthetic experimental-style points (inset: enlarged view near the lowest invariant).

The eutectic reaction L → (Al) + (Si) is calculated at 850 K, with L at x(Si) = 0.121.

Table 1 compares the calculated temperatures with experimental values from the literature [3–5]; in this synthetic document those values and references are fictitious.

Table 1. Invariant reactions and congruent points, calculated and experimental (Al–Si).

Reaction	Type	T / K		Compositions (x(Si))		
		this work	exp.	phase 1	phase 2	phase 3
L → (Al) + (Si)	eutectic	850.2	846.2 [3]	0.121	0.017	1.000

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.
Experimental temperatures are fictitious values attributed to invented references [3–5] for testing evidence-class extraction; they are not data.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Al–Si).

Phase	Pearson	SB	Prototype	T min (K)	T max (K)	Range (x(Si))	T (K)
(Al)	cF4	A1	Cu	< 300[a]	933.5	0.000–0.017	850
(Si)	cF8	A4	C (diamond)	< 300[a]	1687.0	1.000	300

Structure labels follow the TDB phase names (– where the name does not identify a structure).
Stability range: temperatures between which the phase is stable in some part of the diagram.
[a] Stable at the lowest calculated temperature.

(Al) is stable up to 933 K. The solubility of Si in (Al) reaches a mole fraction of 0.017 at 850 K.

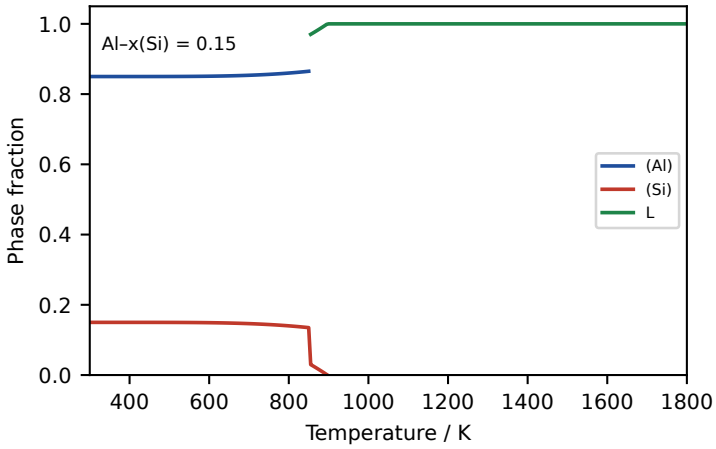


Fig. 2. Calculated equilibrium phase fractions versus temperature for Al–x(Si) = 0.15.

4. Thermodynamic properties

At 1750 K the integral mixing enthalpy of the liquid has its minimum, −3.46 kJ/mol, at x(Si) = 0.520. For the equiatomic liquid at 1750 K, the Gibbs energy of mixing is −13346 J/mol.

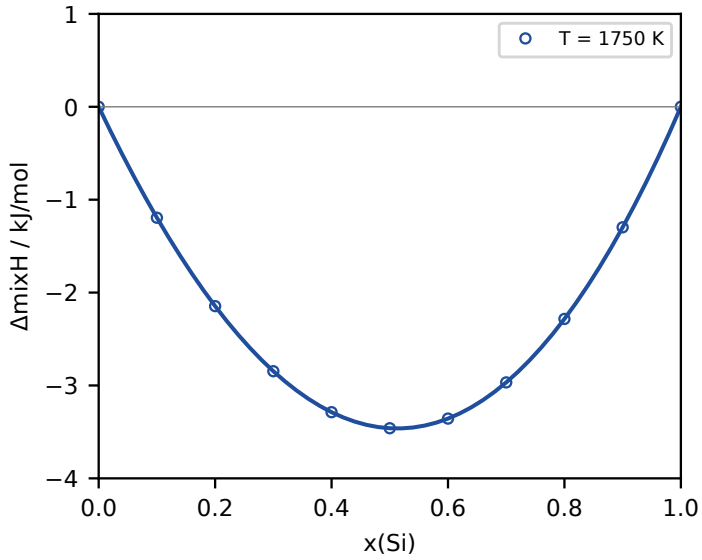


Fig. 3. Calculated mixing enthalpy of the liquid at 1750 K.

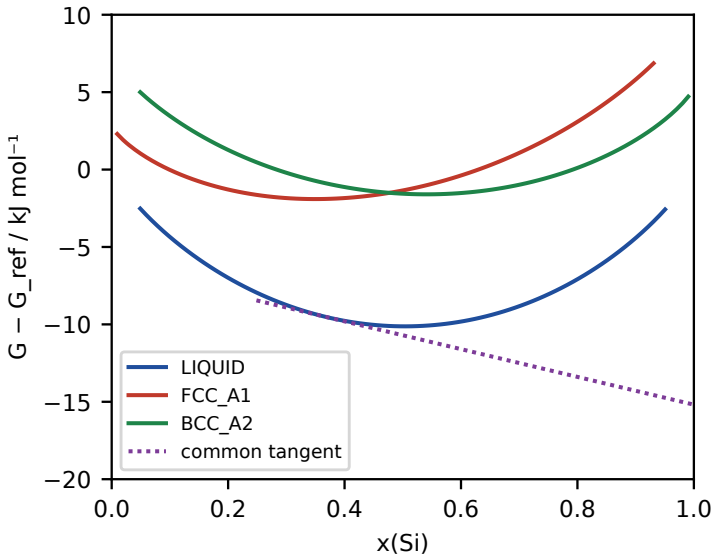


Fig. 4. Calculated Gibbs energy curves at 1180 K.

Table 3. Redlich–Kister parameters of the liquid (Al–Si).

	Liquid: L(ν) = a + b·T + c·T·ln T		
ν	**a**	**b**	**c**
L0	−13842.105	0.45733	0.0
L1	717.10348	−0.2012	0.0

Values as printed in the TDB file.

Table 4. Integral and partial mixing properties of the liquid at 1750 K (Al–Si).

at.% Si	ΔmixH / J mol⁻¹	ΔmixG / J mol⁻¹	a(Al)	a(Si)
10	−1194	−5878	0.8926	0.0490
30	−2847	−11596	0.6484	0.1929
50	−3461	−13346	0.4021	0.3971
70	−2967	−11658	0.1938	0.6431
90	−1297	−5930	0.0478	0.8914

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] Y. Tang, Y. Du, L. Zhang, X. Yuan, G. Kaptay, Thermochim. Acta, 527, 131-42(2012).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.